Waves

Wave

A wave is a disturbance that transfers energy from one point to another.
It is the disturbance which travels, not the particles.
particles only oscillates.

>> wave is a form of energy transfer and momentum transfer.

>> Disturbance may be in the form of :-

(i) displacement e.g.- waves on string
(ii) pressure e.g sound waves
(iii) desity
(iv) Electric and magnetic Field eg:- electro magnetic waves (EMW)

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→ Types of wave

① Mechanical Wave

• Requires a medium to travel.

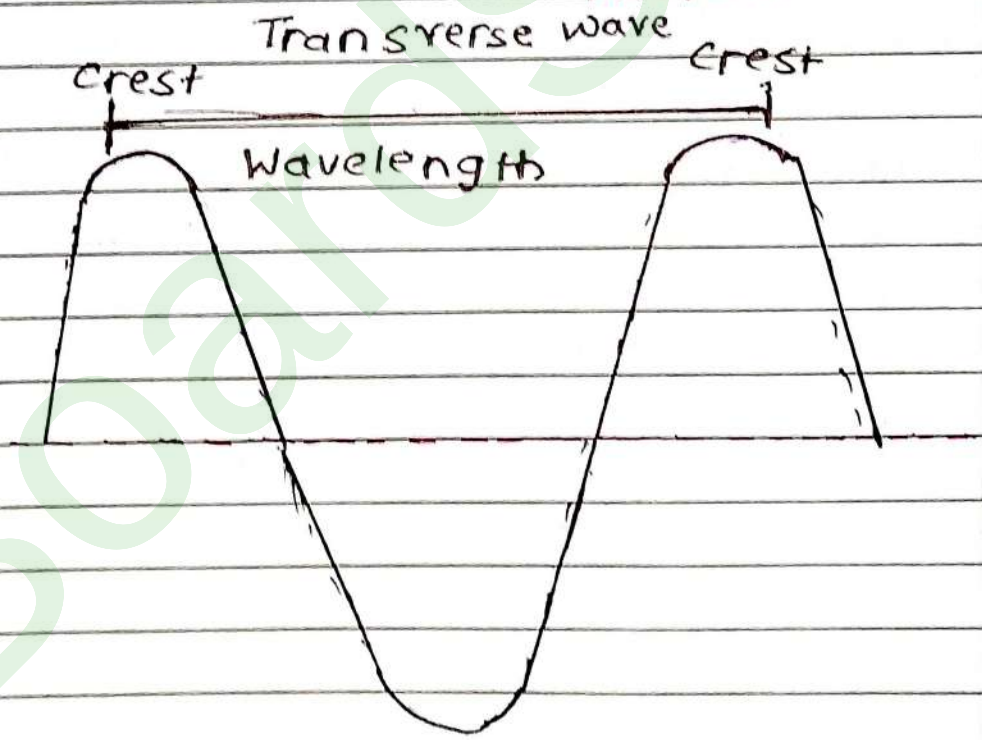
e.g Sound waves, waves on string & in air column.

② Non-mechanical wave

• Do not require a medium to travel

e.g. Light waves (EMW)

TRANSVERSE AND LONGITUDINAL WAVES



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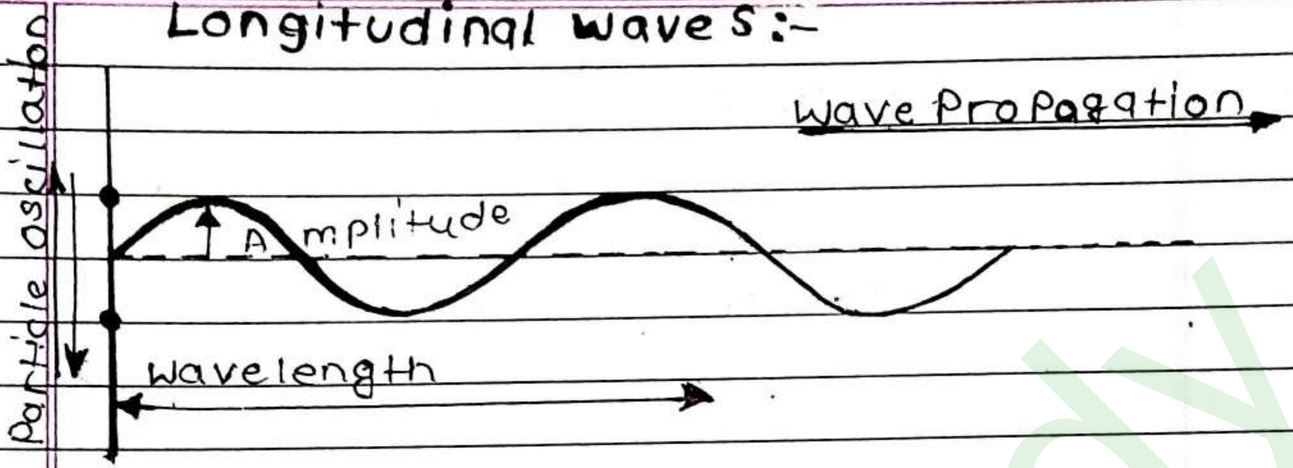
Longitudinal waves:-



Aspect	Transverse Waves	Longitudinal waves
Particle motion	Perpendicular to the direction of wave propagation.	Parallel to the direction of wave propagation.
wave components	crest (high points) and troughs (law points	compression (high pressure and rare-Factions (low pressure)

Medium Requirement	can propagate through solid liquid and vaccum (in the case of electro-magnetic wave).	Typically requires a medium (solid, liquid or gas) for propagation.
Examples	Light waves, wave on a string surface water waves	sound wave in air seismic P-waves, waves in a slinky.
visual Representation	can be easily visualized as peaks and valleys on a Graph.	often represented as compression and expansions along a line
Propagation speed	The speed depends on the tension and mass per unit length in a medium (For mechanical wave like waves on a string.	Speed depends on the medium's density and elasticity; typically Faster in solids, Slower in gases.
medium Requirment	can Travel through a vaccum (in the case of electro-magnetic waves).	Required a material medium (cannot travel through a vaccum)

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General Equation of wave

- If some quantity 'y' (displacement, pressure, density or EMW)
  Let $y \rightarrow$ displ. of particle from mean position.

$$y = f(t)$$
$$y \; f(x)$$
$$y = F(x,t) = \boxed{y = f(ax \pm bt)}$$

This is the equation of wave if :-

① $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \quad (v^2 \neq 0)$

② y is defined for all 'x' and 't'

velocity of wave $= \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{b}{a}$

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Plane Progressive (travelling) Harmonic waves.

Syllabus

>> waves in which all particles are doing S.H.M

Equation of Travelling Harmonic wave

$$\boxed{y = A\sin(kx \pm \omega t)}$$

y = displace of particle from mean position.

A - Amplitude

k = wave number $\left(k = \frac{2\pi}{\lambda}\right)$

x = wave travels along x

ω = Angular frequency.

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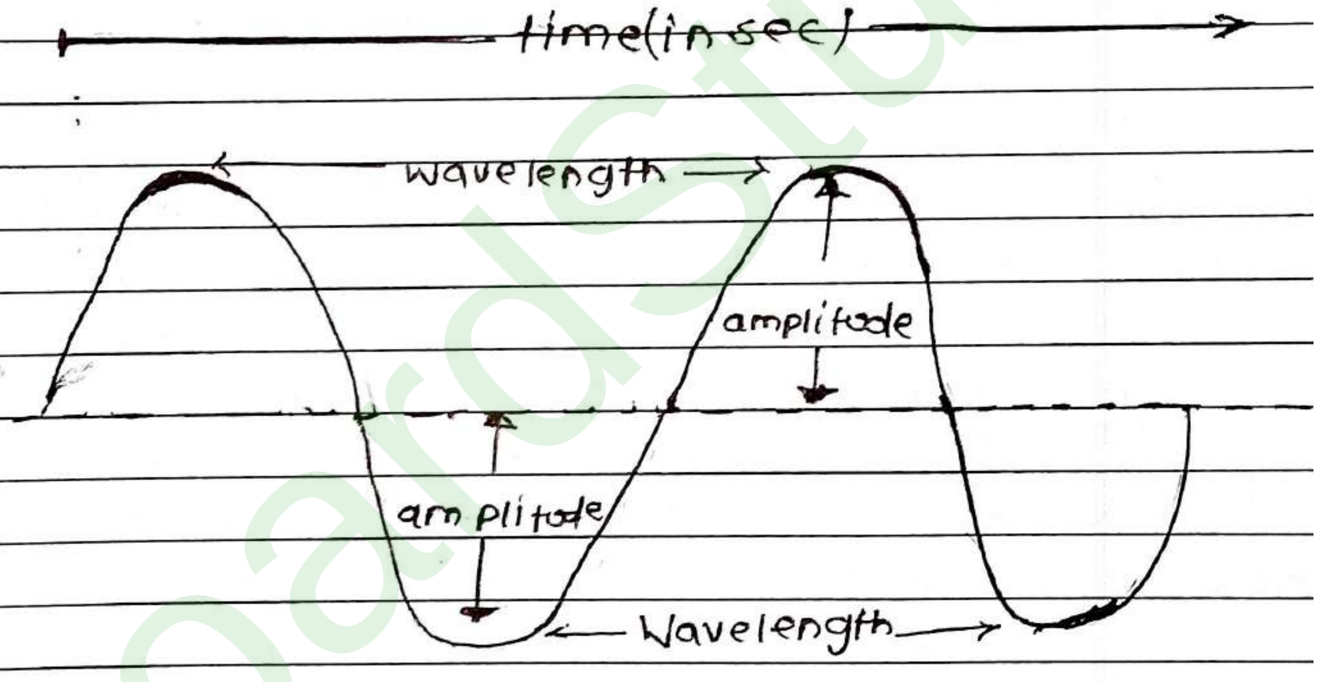
Explanation of the terms

- Amplitude A - The Amplitude is maximom displacement of the particles from their equiibrium position It represents the height of the wave crest or the depth of the through.



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Wave Number k :- The wave number k gives the number of wavelengths per unit distance. It is related to the wavelength λ by the relation $k = \frac{2\pi}{\lambda}$.

Angular frequency ω :- The angular frequency ω is related to the time period 'T' and the frequeny of the wave. It represents how rapid

Phase $(kx - \omega t + \phi)$:
The phase of the wave idicates the state of the oscillation at a particular position and time. It determines the position of the wave within it's cycle. The term $kx - \omega t$ shows the wave's progression in space and time. The phase constant shifts the wave along the x- axis.

Wave speed (v):
How fast the wave moves through the medium, calculated by the formula:

$$v = f\lambda$$

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- Velocity of wave $= \frac{\omega}{K} = \frac{2\pi f}{\frac{2\pi}{\lambda}} \Rightarrow \boxed{V = f\lambda}$

- Direction of Harmonic wave

$\left.\begin{array}{l} \omega t + Kx \\ -\omega t - Kx \end{array}\right\}$ same sign $\rightarrow$ $-$ve direction

$\left.\begin{array}{l} \omega t - kx \\ Kx - \omega t \end{array}\right\}$ opposite sign $\rightarrow$ $+$ ve direction.

V - depends on medium
f - depends on source
λ - depends on medium

Phase and phase difference ($\Delta\phi$)

$y = A\sin(\underbrace{\omega t + Kx + \phi}_{\text{phase}})$

$\cos\theta = \sin(\theta + 90^\circ)$
$-\cos\theta = \sin(\theta - 90^\circ)$

$y_1 = A_1 \sin(\underbrace{K_1 x + \omega_1 t}_{\phi_1})$

$y_2 = A_2 \sin(\underbrace{K_2 x + \omega_2 t}_{\phi_2})$

$\Delta\phi = \phi_2 - \phi_1 = (K_2 x + \omega_2 t) - (K_1 x + \omega_1 t)$

Que. $y_1 = a\sin(\omega t + kx + 0.57)$ m
$y_2 = a\cos(\omega t + kx)$ m, Find phase difference

Sol- $y_2 = a\cos(\omega t + kx) = a\sin(\omega t + kx + \pi/2)$

$\Delta\phi = \phi_2 - \phi_1$

$= \frac{\pi}{2} - 0.57 = \frac{3.14}{2} - 0.57$

$= 1.57 - 0.57$

$= 1$ radian

Wave velocity and particle velocity

$y = A\sin(kx - \omega t)$ (Let -ve x- direction)

wave velocity $\Rightarrow V = \frac{\text{coefficient of } t}{\text{coefficient of } x} = \frac{\omega}{k} \Rightarrow \boxed{V = f\lambda}$

particle velocity $\Rightarrow$ y $\Rightarrow$ displacement of particle.

$\boxed{V_p = \frac{dy}{dt}}$ $\boxed{(V_p)_{max} = \omega A}$ → Particle is doing SHM

Ques: $y = 3in\frac{\pi}{2}(50t - x)$. Find the maximum particle velocity to the wave velocity is ?

Solv:

$$y = 3\sin\left(\frac{50\pi}{2}t - \frac{\pi x}{2}\right)$$

$$y = A\sin(\omega t - kx)$$

$$\frac{(V_p)_{max}}{V_{wave}} = \frac{\omega A}{\frac{50}{1}} = \frac{\frac{\pi \times \cancel{50} \times 3}{2}}{\cancel{50}} = \frac{3\pi}{2} \text{ Ans}$$

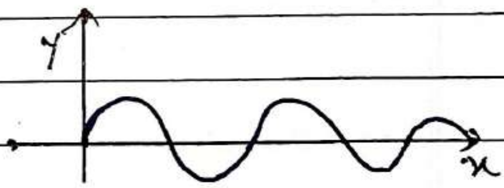
- Relation b/w wave velocity and particle velocity

$$\boxed{V_p = -v\left(\frac{\partial y}{\partial x}\right)}$$

velocity of particle

Velocity of wave

slope of y-x



slope = +ve

Slope = -ve.

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Acceleration of particle

$y = A \sin (kx - \omega t)$

$V_p = \frac{dy}{dt} - \omega A \cos (kx - \omega t)$

$a_p = \frac{dv_p}{dt} = -\omega^2 \underbrace{A \sin (kx - \omega t)}_{y}$

$\boxed{a_p = -\omega^2 y}$

$\boxed{(a_p)_{max} = \omega^2 A}$ ← Particles are in S.H.M extreme position

⇒ a in SHM is always towards Mean position.

Phase difference for same particle at different time.

Let's consider particle at 'x' distance at time t_1,

$y_1 = A \sin (\omega t_1 - kx)$

at time t_2,

$y_2 = A \sin (\omega t_2 - kx)$

$\Delta \phi = \omega t_2 - \omega t_1$

$\Delta \phi = \omega (t_2 - t_1)$

$$\boxed{\Delta \phi = \omega \Delta T}$$

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Phase difference b/w two particle at a given time:-

For two different particles at time 't'
For 1st particle

$$y_1 = A\sin(kx_1 - \omega t)$$

For 2nd particle

$$y_2 = A\sin(kx_2 - \omega t)$$

$$\Delta\phi = kx_2 - kx_1$$

$$\Delta\phi = k(x_2 - x_1)$$

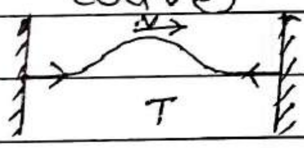
$\Delta\phi \rightarrow$ Phase difference

$$\boxed{\Delta\phi = k\Delta x}$$

$\Delta x \rightarrow$ Path difference.

Speed of wave on string (Transverse wave)

- Depends on medium



$$\boxed{v = \sqrt{\frac{T}{\mu}}}$$

$T \Rightarrow$ tension in string
$\mu =$ Mass per unit length.

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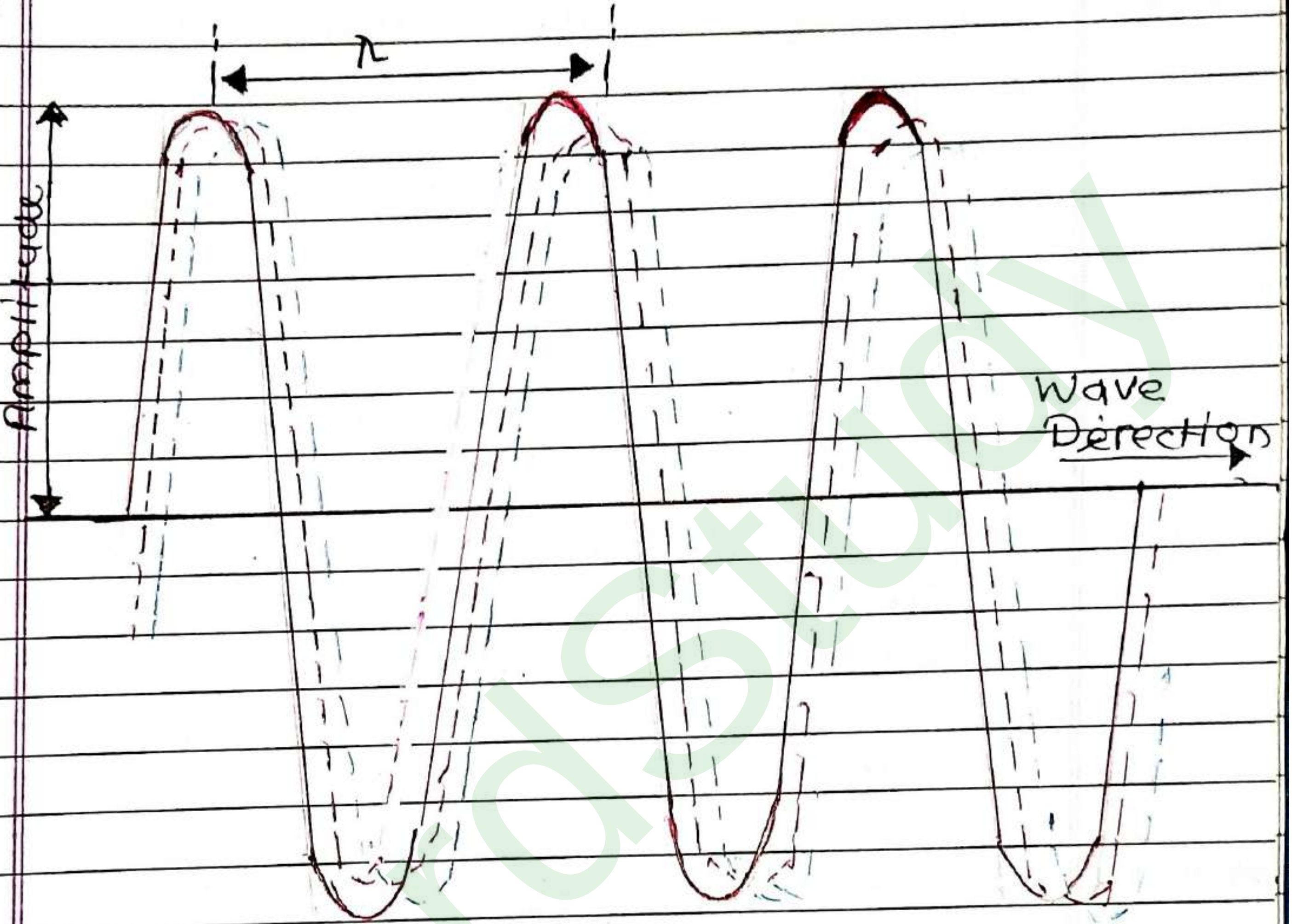
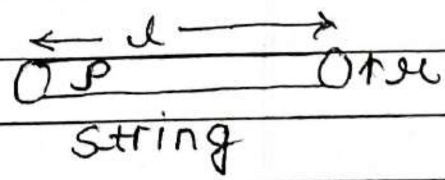


$m = \rho \times V$

$m = \rho = (\pi r^2 l)$

$\frac{m}{l} = \rho \pi r^2$



$$\boxed{V_{(wave)} = \sqrt{\frac{T}{\rho \pi r^2}}}$$

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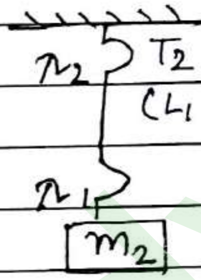
>> A uniform rope of length L and Mass m, hangs vertically from a rigid support



(f, $\mu \rightarrow$ constant)

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 + m_2}{m_2}}$$

Intensity of wave

Intensity $\rightarrow$ Amount of energy passing normally thought a unit area per second.

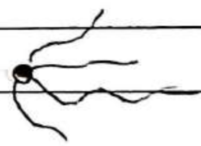
$$I = \frac{E}{At} \quad (\text{unit} \rightarrow \text{watt}/m^2)$$

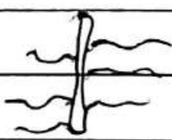
$$\boxed{I = 2\pi^2 f^2 A^2 \rho v}$$

$$I \propto A^2$$

$\rho \rightarrow$ density of medium
$v \rightarrow$ wave velocity.
$A \rightarrow$ Amplitude.

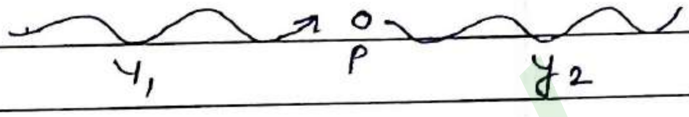
Intensity with distance from source:—

(1) Point source  $I \propto \frac{1}{d_2}$
(d= distance)

(2) Linear source  $I \propto \frac{1}{d}$

Interference of waves

(two waves meets)

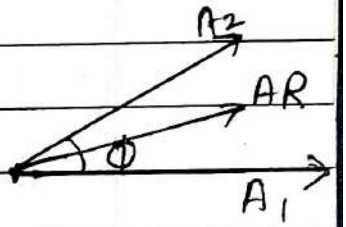


($\vec{y}_{net} = y_1 + y_2$)
(displacement) [Principle of Superposition]

Resultant Amplitude & Intensity

$y_1 = A_1 \sin(\omega t_1 + kx_1)$
$y_2 = A_2 \sin(\omega t_2 + kx_2)$ ϕ } combination of SHM

$y_{net} = y_1 + y_2$



Phase differences (ϕ) $= (\omega t_2 + kx_2) - (\omega t_1 + kx_1)$

$$\boxed{A_R^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$$

$(A_R^2 \propto I_R)$

$$\boxed{I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi}$$

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Constructive Interference	Destructive Interference
wave meet	wave seperate
$I_R, A_R \rightarrow$ maximum	$I_R, A_R \rightarrow$ minimum
$(\cos\phi) \rightarrow$ max. (1)	$(\cos\phi) \rightarrow$ minimum (-1)
$(\phi = 0, 2\pi)$ --- $(2n\pi)$	$(\phi = -1)$ $(\phi = (2n-1)\pi)$
Path difference (Δx)	Path difference (Δx)
$\Delta x = 0, \lambda, 2\lambda$ ---	$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}$ ---
$(\Delta x = n\lambda)$	$[\Delta x = (2n-1)\lambda/2]$
Put $\cos\phi = 1$ in A_R I_R form ula	Put $\cos\phi = -1$
$(A_R)max = A_1 + A_2$	(A_R) min $= A_1 - A_2$
$(I_R)max = (\sqrt{I_1} + \sqrt{I_2})^2$	(I_R) min $= (\sqrt{I_1} - \sqrt{I_2})^2$

$$\frac{I_{max}}{I_{min}} = \left(\frac{A_{max}}{A_{min}}\right)^2$$

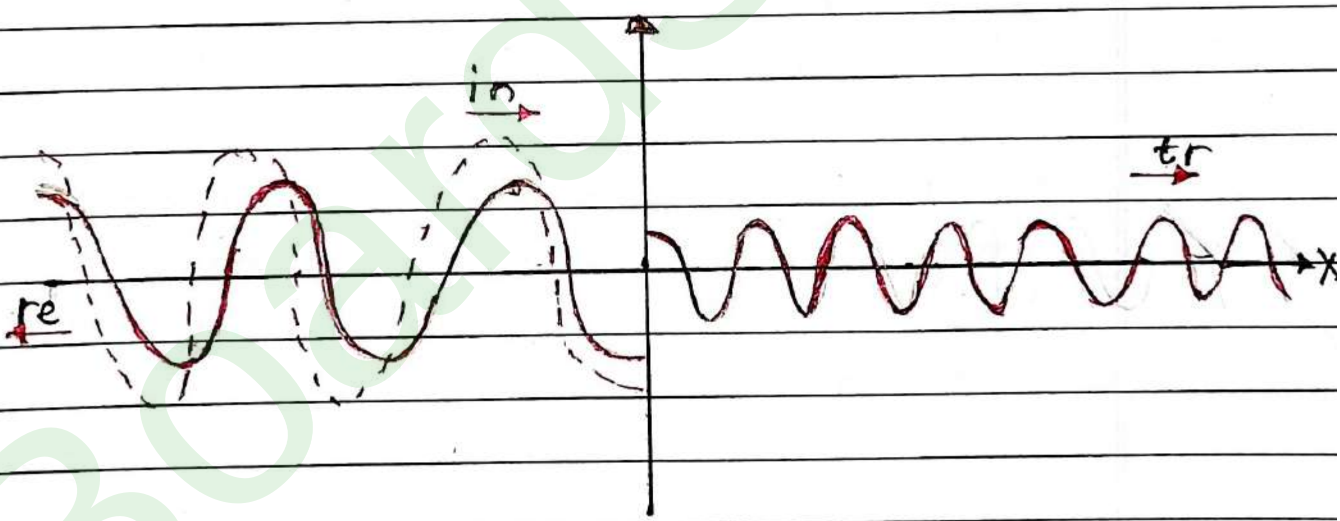
Que The suporposing waves are represented by the following eq^n.

$$y_1 = 5\sin 2\pi (10t - 0.1x)$$
$$y_2 = 10\sin 2\pi (20t - 0.2x)$$

Ratio of intesities $\frac{I_{max}}{I_{min}}$ will be?

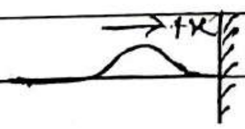
$$\frac{I_{max}}{I_{min}} = \left(\frac{A_{max}}{A_{min}}\right)^2 = \left(\frac{10+5}{5-10}\right)^2 = \frac{15 \times 15}{5 \times 5} = 9$$

	Reflection	Transmission (Refraction)
	(same medium)	(different medium)
Velocity (v)	Same	change
Frequency (f)	Same	Same
Wavelength	Same	change
Phase difference	Rarer → Denser ($\Delta\phi = \pi$) Denser → Rarer ($\Delta\phi = 0$)	$\Delta\phi = 0$



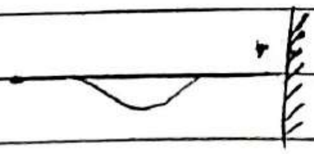
(Denser)

▶ Reflection of wave from fixed end



(No loss of energy)

$y_i = A\sin(kx - \omega t)$



$y_r = -A\sin(kx + \omega t)$

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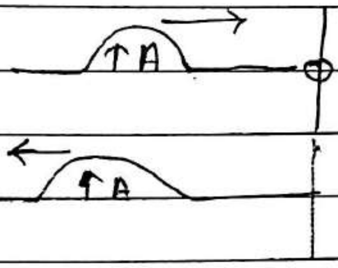
Reflection from free end (Rarer)



$y_i - A\sin(kx - \omega t)$

$y_R = A\sin(kx + \omega t)$

Reflected and transmitted Amplitude

$$A_R = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_i$$

$$A_t = \left(\frac{2v_2}{v_1 + v_2}\right) A_i$$

Standing wave / stationary wave

→ when two harmonic waves of equal frequency and equal Amplitude travelling in opposite direction suprimpose each other, standing wave are formed.

→ No energy transfer.

Conditions for forming standing wave :-

(i) Superposition of two harmonic wave in same medium (v is same)

(ii) $A, f, \omega \rightarrow$ same

(iii) travelling in same direction
$k \rightarrow$ same
Phase difference can be

$\boxed{\text{Equation}}$

e.g.

$y_1 = A\sin(kx - \omega t)$
$y_2 = A\sin(kx + \omega t)$
$y_{net} = y_1 + y_2 = A\sin(kx - \omega t) + A\sin(kx + \omega t)$

$= A[\sin\underbrace{(kx - \omega t)}_{C} + \sin\underbrace{(kx + \omega t)}_{D}]$

Basic maths, $\sin C + \sin D = 2\sin\left[\frac{C+D}{2}\right]\cos\left[\frac{C-D}{2}\right]$

$y = 2A\sin(kx)\cos(-\omega t)$

$\boxed{y = 2A\sin(kx)\cos(\omega t)}$

$\leftarrow$ No $(kx \pm \omega t)$

can be in other forms also.

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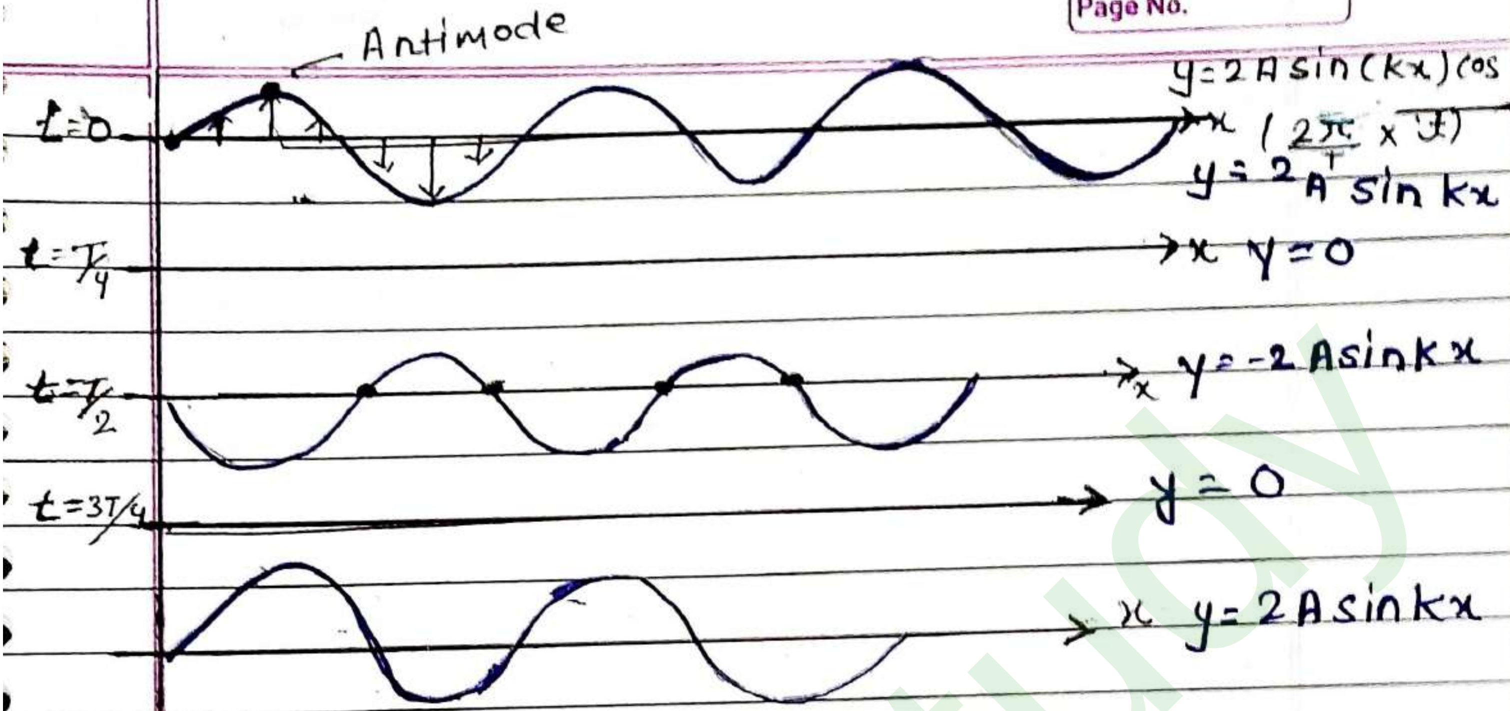

Antimode
t=0
y=2A sin(kx) cos(2π/T × t)
y=2A sin kx
t=T/4
y=0
t=T/2
y=-2A sin kx
t=3T/4
y=0
y=2A sin kx
x

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① All particles execute SHM with different Amplitudes.

② Particles which have max. amplitude → Antinodes

Particles which have min. amplitude → Nodes ($A = 0$)

③ Some particles can never move from mean position → Nodes

④ Particle of one segment are all in same phase ($\Delta\phi = 0$)

But with adjacent segment in opp phase ($\Delta\phi = \pi$)

⑤ All particles reach their maximum displacement simultaneously.

⑥ Twice in 1 time period, all particles reach their mean position simultaneously

⑦ No energy or momentum is transferred in stationary waves

⑧ velocity of standing wave is not zero, as expected.

→ Distance b/w 2 consecutive nodes = $\lambda/2$

→ Distance b/w 2 consecutive antinodes = $\lambda/2$

→ Distance b/w consecutive node and antinode = $\lambda/4$.

• Velocity of standing wave = velocity of wave interfering to form stationary wave.

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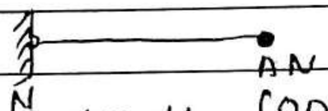
Comparison of progressive and standing wave

Progressive wave (Travelling wave)	Standing wave
1. These waves travel in a medium with definite velocity.	1. Do not travel and remain confined between two boundries in medium
2. These waves transmit energy in the medium	2. These waves do not transmit energy in the medium.
3. These phase of vibration varies continuously from particle to particle.	3. The phase of all particles in b/w two nodes is always same, But particles of two adjacent nodes differ in phase by 180°.
4. No particle of medium is permanently at Rest.	4. Particles at node are permanently at Rest.
5. All particles donot attain the max. displacement position simultaneously	5. All particles attain the maximum displacement simultaneously.

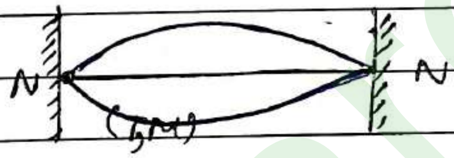
6. All particle of the medium vibrate and amplitude of vibration is same.	6. The amplitude of vibration changes from particle to particle. The amplitude is zero for all at nodes and maximum at antinodes



▶ Normal Modes of strings

1. Fundamental mode or tone / principal Tone / I^{st} Harmonic

1 segment.



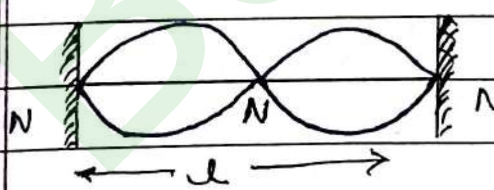
$\lambda/2 = l$

$[\lambda = 2l]$

$\left[f_1 = \frac{V}{2l}\right]$ ← Fundamental Frequency

2. First overtone / second harmonic

2 segment



$\frac{2\lambda}{2} = l \Rightarrow \lambda = \frac{2l}{2}$

$f = \frac{V}{\lambda} = \frac{V}{\frac{2l}{2}} = 2\left(\frac{V}{2l}\right) \Rightarrow \left[f_2 = 2f_1\right]$

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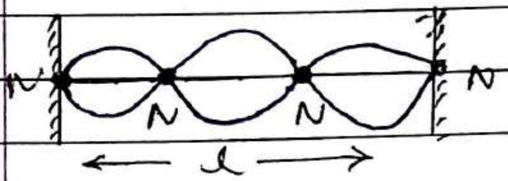
3. Second overtone / Third Harmonic
↑ 3 segment



$\frac{3\lambda}{2} = l \Rightarrow \lambda = \frac{2l}{3}$

$f = \frac{v}{\lambda} = \frac{v}{2l/3} = 3\left(\frac{v}{2l}\right) \Rightarrow \left[f_3 = 3f_1\right]$

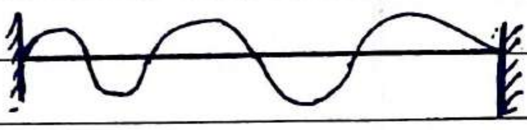
No of Node = (no of segement +1)
No of Antinode = no of segment.

n^{th} overtone $\Rightarrow$ $(n+1)^{th}$ harmonic, segments, antinodes.
$(n+1) f_1$
$(n+2)$ nodes.

→ Resonant frequencies → Harmonics
(Natural frequencies)

Qus Standing waves are produced in a 10 m long stretched string. If the string vibrates in 5 segments and the wave velocity in 20 m/s the frequncy is?

Solve $f = \frac{v}{\lambda} = \frac{20}{4} = 5\,Hz$



$\frac{5\lambda}{2} = 10$

$(\lambda = 4)$

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Ques A string is streched b/w Fixed points Separeted by 75 cm. It is observed to have resonant frequencies of 420 Hz. and 315 Hz. There are no other resonant frequencies b/w these two. The lowest resonant frequencies for the string is?

Solve Resonant frequencies → Harmonics
Lowest resonant frequency → Fundametal tone.

315 Hz → N^{th} harmonic → nf_1
420 Hz → $(n+1)^{th}$ harmonics → $(n+1)f_1$
Fundamental / I^{st} harmonic ⇒ f_1

$$420 = nf_1 + f_1$$
$$420 = 315 + f \Rightarrow f_1 = 105 \text{ Hz}.$$

Ques when a string is devided into three segmets of length l_1, l_2 and l_3. the Fundamental frequeny of these three segements are

Solve f_1, f_2, f_3 The original fundalmental Frequency (v) of string is

$f_1 \; l_1 \; | \; f_2 \; l_2 \; | \; f_3 \; l_3$

$l = l_1 + l_2 + l_3$

$l = l_1 + l_2 + l_3$

$$\frac{V}{2f} = \frac{V}{2f_1} + \frac{V}{2f_2} + \frac{V}{2f_3}$$

$$f_1 = \frac{V}{2l_1}$$

$$f_2 = \frac{V}{2l_2}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$f_3 = \frac{V}{2l_3}$$

SOUND WAVES

- Sound wave is a longitudinal wave.
- Infrasonic wave :- longitudinal wave with $f < 20 Hz$.
- Audible wave (sound wave) - Longitudinal wave with $f = 20 Hz$ to $20,000 Hz$.
- Ultrasonic wave - Longitudinal wave with $f > 20,000 Hz$.

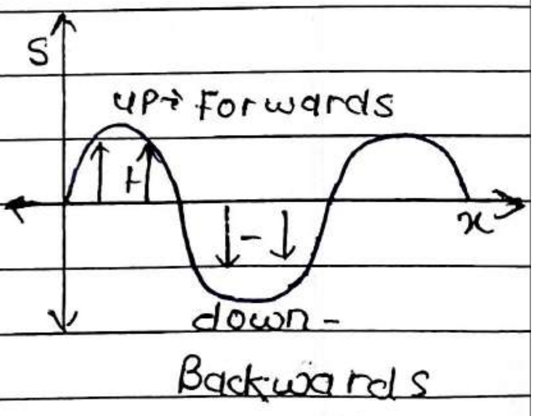
It have max. energy and can travel long distance.

→ Sound wave travels due to pressure and density variation.

Compression → $\left.\begin{matrix}\text{Pressure}\\ \text{Density}\end{matrix}\right\}$ maximum

Rarefaction → minimum.

► Equations of sound waves



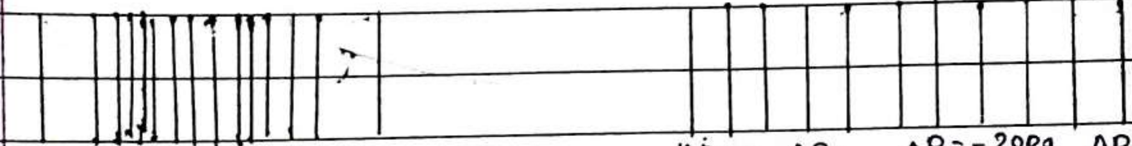
1. Displacement wave eq^n

$\boxed{S = A\sin(kx - \omega t)}$

↓ disp. of particle (x-direction)

↓ wave travels in +x direction

2. Pressure wave Equation



$\Delta P = 0$ $\quad P = +20$ $\quad P = +10$ $\quad \Delta P = 0$ $\qquad \Delta P = 0$ $\quad \Delta P = -10Pa$ $\quad \Delta P = -20Pa$ $\quad \Delta P = 0$

compression $\qquad\qquad$ Rarefaction

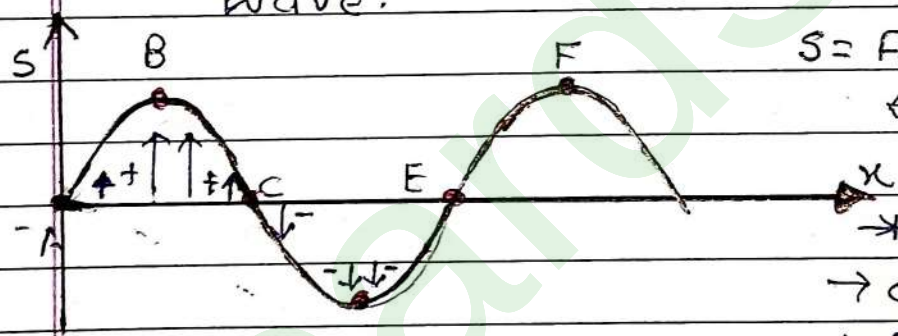
→ ΔP varies harmonically

→ ordinary pressure = 1 atm = 10^5 Pa.

($\Delta P \rightarrow$ max.)

→ Displacement node is the pressure antinode and displ.. antinode is pressure node, in sound wave.

($\Delta P = 0$)



$S = A \sin(kx - \omega t)$

$\leftarrow A \mapsto$ Rarefaction

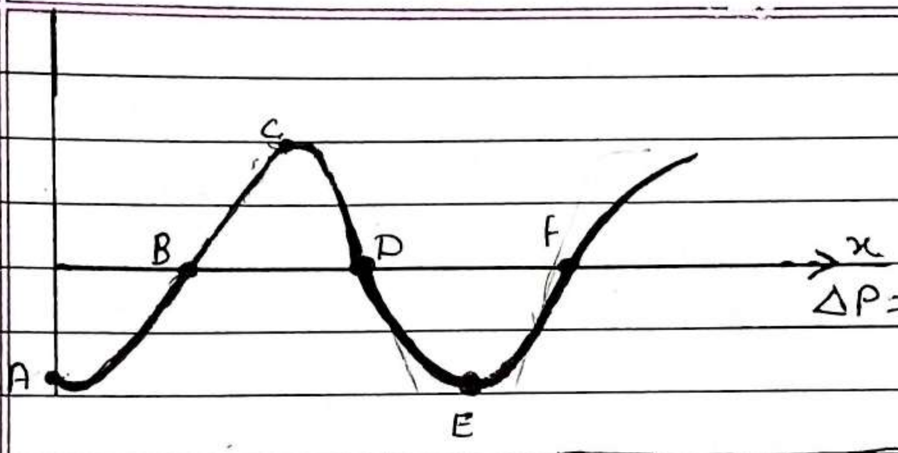
$\Delta P \rightarrow$ min (−ve)

→ $B \mapsto \Delta P = 0$

→ C ← compression

ΔP = max (+ve)

$\leftarrow D \mapsto$ $\Delta P = 0$



$\Delta P = -\Delta P_0 \cos(kx - \omega t)$

Equation, $\boxed{\Delta P = -\Delta P_0 \cos(kx-\omega t)}$
$\boxed{= \Delta P_0 \sin(kx - \omega t - \pi/2)}$

- ϕ (phase difference) b/w displace. wave and pressure wave is $\pi/2$.

- $\boxed{\Delta P_0 = BkA}$ β = Bulk modulas
  k = wave no ($2\pi/\lambda$)

▶ Speed of sound [Longitudinal wave]

(I) Speed of sound wave

$\boxed{v = \sqrt{\frac{E}{\rho}}}$ $E \rightarrow$ coefficient of elasticity
$\rho \rightarrow$ density of medium

$V_{solid} > V_{liquids} > V_{gas}$

⇒ In liquids / Gases, $E = \beta$

$\left(v = \sqrt{\frac{\beta}{\rho}}\right)$

⇒ In solids, $E = Y$

$\left(v = \sqrt{\frac{Y}{\rho}}\right)$

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▸ Speed of sound in air (According to Newton)

- According to Newton, sound travels in air isothermally, i.e at constant temperature.

- Bulk modulus for isothermal pressure,
$\beta_{iso} = P$.

$$V = \sqrt{\frac{\beta}{\rho}} = \sqrt{\frac{P_{air}}{\rho_{air}}} \approx 280 \text{ m/s}$$

but experimentaly,
$V \approx 332$ m/s

▸ Laplace correction

- A/Q to Laplace, sound travels in air adiabatically $(\Delta Q = 0)$

Bulk modulus for adiabatic process,
$\beta_{ad} = \gamma P$

- $$\left[V = \sqrt{\frac{\beta}{\rho}} = \sqrt{\frac{\gamma P}{\rho}}\right] \approx 332 \text{ m/s}$$

$\gamma_{air} = 1.4$ (diatomic)
$P_{air} = (10^5$ Pa)
$\rho_{air} = 1.29$ kg/m^3

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▸ In terms of Absolute temperature

$$\boxed{v = \sqrt{\frac{\gamma R T}{M}}}$$

Density of Ideal Gas

$$\rho = \frac{PM}{RT} \Rightarrow \frac{P}{\rho} = \frac{RT}{M}$$

R = 8.314 J/mol K

T → in K

M → molar mass in kg.

γ → 1 + 2/f

* If T is in °C ($t \le 25°C$)

($v_{t°C} = v_{0°C} + 0.61 \times t$)

* Factors affecting speed of sound in air.

→ Effect of temperature, $v \propto \sqrt{T}$

If it is contant, no effect of presure

Effet of Humidity, $\rho_{moist\ air} < \rho_{dry\ air}$

$$v \propto \sqrt{\frac{1}{\rho}}$$

(moist air contains water vapour → lighter)

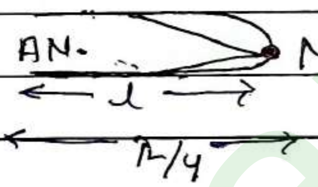
ORGAN PIPE

→ ORGAN PIPE :- Pipe which is made up of metal, wood & Sometime with glass.

1. Open organ Pipe :- Both ends are open. Both ends have disp. AN.

Closed organ pipe :- open end is closed.
- → closed end - Displacement Node
- → open end - Dispacemet anti node.

► Fundamental tone / Ist Harmonic

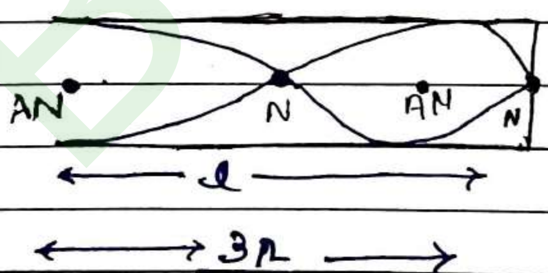


$$\frac{\lambda}{4} = l$$

$$\lambda = 4l$$

$$f = \frac{v}{\lambda} \Rightarrow \boxed{f_1 = \frac{v}{4l}}$$

First overtone / 3rd Harmonic



$$\frac{3\lambda}{4} = l$$

$$\lambda = 4l/3$$

$$f = \frac{v}{\lambda} = \frac{3v}{4l} \Rightarrow \left[f_{I^{st}\ \text{overtone}} = 3f_1 \right]$$

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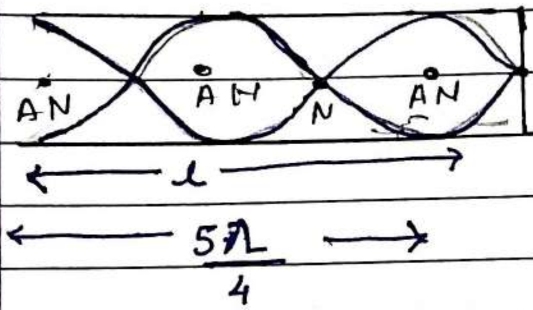
Seond overtone / 5^{th} harmonic



$$\frac{5\lambda}{4} = l \rightarrow \lambda = \frac{4l}{5}$$

$$f = \frac{v}{\lambda} = \frac{5v}{4l}$$

$$\Rightarrow \left[f_{2nd\ overtone} = 5f_1 \right]$$

$\Rightarrow$ Closed organ pipe has only odd Harmonics.

First overtone $= 3f_1$
f_{2d} overtone $= 5f_1$
f_{nth} overtone $(2n+1) f_1$

Que The number of possible natural oscillation of Air column in the pipe closed at one end of length 85cm whose frequencies lie below 1250 Hz are? (velocity of sounds = 340 m/s)

Sol ~~n^{th} harmonic $\rightarrow$ 220 Hz $= nf_1$~~
~~$(n+2)^{th}$ harmonic $\rightarrow$ 260 $= (n+2) f_1$~~
~~$nf_1 + 2f_1 = 260$~~
~~$220 + 2f_1 = 260$~~
~~$2f_1 =$~~

$$f_1 = \frac{v}{4l} = \frac{340 \times 100}{4 \times 85} = 100\ Hz \Rightarrow$$

F	3f	5f	7f	9f	11f
100	300	500	700	900	1100

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Ques The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. what is the Fundamental frequency of the system?

Solve

n^{th} Harmonic $\rightarrow 220\,Hz = nf_1$

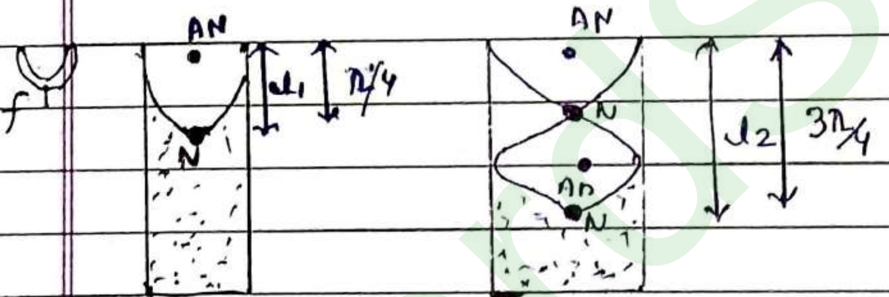
$(n+2)^{th}$ harmonic $\rightarrow 260 = (n+2)f_1$

$= nf_1 + 2f_1 = 260$

$220 + 2f_1 = 260$

$2f_1 = 40 = f_1 = 20\,Hz$.

Resonance in air column (closed organ pipe)



$\Rightarrow$ Resonance – when frequency of external source of matches with natural force of object.

$l \rightarrow$ length of air column

$l_1 : l_2 : l_3 : l_4$

$1 : 3 : 5 : 7$

* Speed of sound from resonance table

$$v = 2f(l_2 - l_1)$$

where l_1 and l_2 are successive resonance length.

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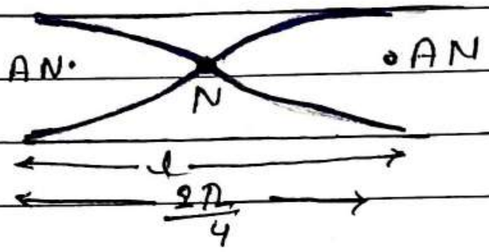
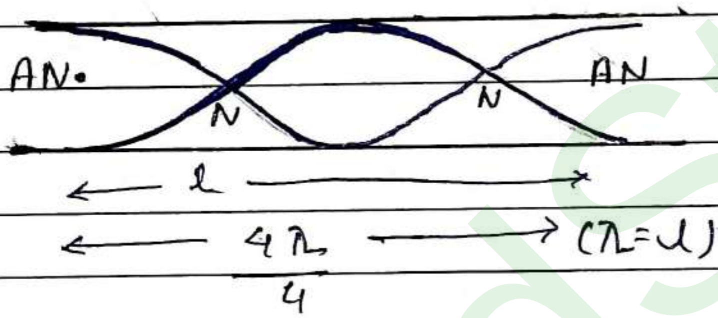
Open organ pipe

1. Fundamental tone / Ist Harmonic



$F = \frac{V}{\lambda} = \frac{V}{2l} = \left[F_1 = \frac{V}{2l} \right]$

2. First overtone / II harmonic



$f = \frac{V}{\lambda} = \frac{V}{l} \Rightarrow f = \frac{2V}{2l}$

$f = 2\left(\frac{V}{2l}\right) = 2f_1$

$\left[f_{1st} \; 2f_1 \right]$ (overtone)

→ Open oxygen pipe has both odd and even harmonics.

Harmonic in open oxygen pipe

① Fundamental tone / Ist harmonic, $F_1 = \frac{V}{2l}$

② First overtone $= 2F_1$
F_{2nd} overtone $3F_1$
F_{3rd} overtone $= 4F_1$
F_{nt} overtone $= (n+1)f_1$

Que The Fundamental Frequency in an open organ pipe is equal to the third Harmonic of a closed organ pipe If the length of the closed organ pipe is 20cm. the length of the open organ pipe is?

F_1 open = F_{3rd} harmonic closed

$$\frac{v}{2l_0} = 3f_1' \text{ closed}$$

$$\frac{v}{2l_0} = \frac{3v}{4l_c} \Rightarrow \frac{1}{2\times l_0} = \frac{3}{4\times 20} \Rightarrow l = 13.2\text{ cm.}$$

BEATS

→ The interference or superposition of two sound waves having small difference in frequency is called beats.

Standing waves	Beats
① Frequency is same	① Frequency is slightly different.
② Travelling in opp. direction	② Travelling in same direction.

→ Destructive Interference - minima (waning) ⇒ $A_R = A_1 - A_2$

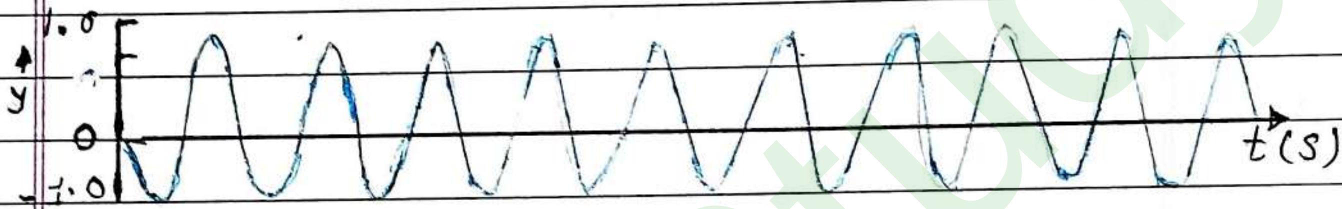
→ constructive Interference - maxima (waxing) = $A_R = A_1 + A_2$

◆ 1 Beat = 1 maxima + 1 minima

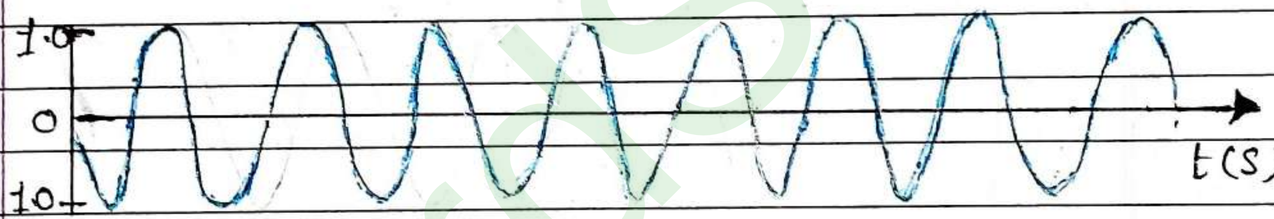
▶ No. of Beats/Beat Frequency = No. of beats per second
No. of maximas/minimas per second.

$$\boxed{\text{Beat Frequency} = |f_2 - F_1|}$$

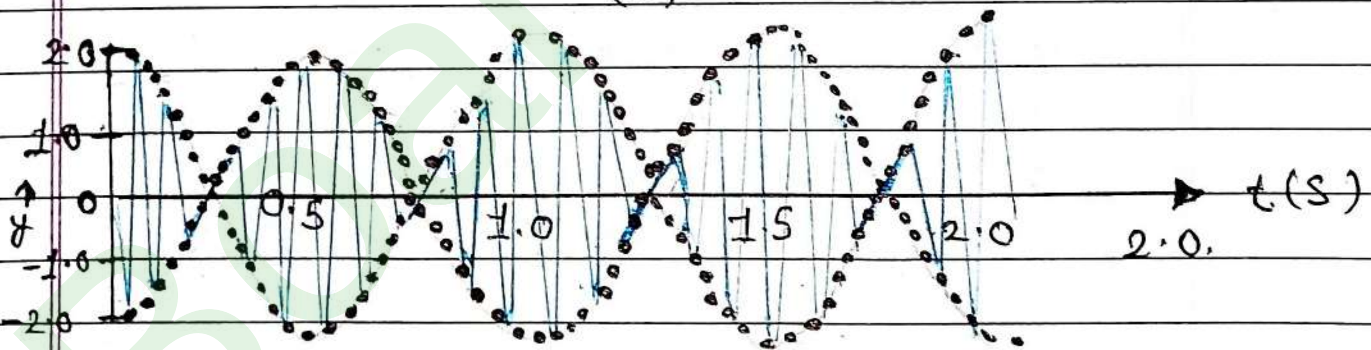
Time period of beat $(T) = \frac{1}{|F_2 - F_1|}$



(a)



(b)



(c)

Superposition of two harmonic waves, one of frequency 11 Hz (a) and the other of frequency 9 Hz (b), giving rise to beats of frequency 2 Hz, as shown in (c).

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Ques A tunning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per sec when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was?

Fork	Piano	Beats
512	516 / 508	4
512	514 510	2

Ans 508 Hz.

Qu- In a guitar two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. when tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz. the original frequency of B will be?

Solve

A	B	Beats
530	536 / 524	6
530	537 / 523	7

Ans = 524 Hz

Tuning fork

① waxing / loading ⇒ $f \downarrow$

② Scrapped ⇒ $f \uparrow$